

Birla Institute of Technology & Science, Pilani

Online B.Sc. Computer Science Programme

Second Semester 2022-2023

Comprehensive Exam

(Slot-1)

Course No. : BCS ZC233
Course Title : Probability & Statistics
Nature of Exam : Open Book
Weightage : 50%
Duration : 2 Hours 30 Minutes
Date of Exam : 08th October 2023

No. of Pages = 2
No. of Questions = 5

Q. No	Question	Marks												
Q.1.a	In a City, 40% of the people use UBER service to go to their offices daily, 70% use OLA service and 30% use both. A person is randomly selected. Then find the probability that he a) uses UBER service or OLA service b) uses OLA service given that he is regularly using UBER service c) uses exactly one Service	5 M												
Q.1.b	An investor made 40% of investment in stocks, 20% in mutual funds and the rest in his personal business. There are 20%, 10% and 15% of chances of obtaining profit in stocks, mutual funds and personal business respectively. a) What is the probability that the investor gets profit? b) Given that the investor has got some profit, what are the probabilities that he got it from stocks, mutual funds and business?	5 M												
Q.2	Consider random variable X with the distribution given as <table><tr><td>x</td><td>0.2</td><td>0.4</td><td>0.5</td><td>0.8</td><td>1</td></tr><tr><td>p(x)</td><td>0.1</td><td>0.1</td><td>0.2</td><td>0.3</td><td>0.3</td></tr></table> a) Validate the distribution b) Find the mean c) Find Variance d) Find $P(X < 0.8)$ e) Find $P(0.2 < X < 0.8)$	x	0.2	0.4	0.5	0.8	1	p(x)	0.1	0.1	0.2	0.3	0.3	10 M
x	0.2	0.4	0.5	0.8	1									
p(x)	0.1	0.1	0.2	0.3	0.3									
Q.3.a	If $P(A) = 1/2$, $P(B) = 1/3$ and $P(A \cap B) = 1/5$ then find a). $P(A \cap B)$ b). $P(A^c \cap B)$ c). $P(A \cap B^c)$	5 M												

	d). $P (A^c \cap B^c)$ e). $P (A^c \cup B^c)$													
Q.3.b	Let X be a random variable which follows normal distribution with mean 25 and standard deviation 3 then find a) $P(22 < X < 28)$ b) $P(X > 28)$ c) $P(X < 28)$ d) $P(X = 26)$	5 M												
Q.4.a	A company claims that mean life of their product is 120 months with Standard deviation 5 months. Sampling is done to validate this claim. Sample of size 40 is having mean life 125 months with standard deviation 2 months. Validate the claim at 1 % level of significance.	5 M												
Q.4.b	It is observed that 70 % of the people in a city prefer dinner at restaurant during weekends. We will validate this observation with the help of sampling at 5% level of significance. 175 persons out of 250 regularly goes out to have dinner at restaurants during weekends.	5 M												
Q.5.a	Find Co-variance of the following data and interpret the result <table border="1"><tr><td>X</td><td>10</td><td>15</td><td>12</td><td>16</td><td>18</td></tr><tr><td>Y</td><td>5</td><td>8</td><td>6</td><td>7</td><td>9</td></tr></table>	X	10	15	12	16	18	Y	5	8	6	7	9	5 M
X	10	15	12	16	18									
Y	5	8	6	7	9									
Q.5.b	Consider the following data and predict y when x = 10 <table border="1"><tr><td>X</td><td>1</td><td>3</td><td>5</td><td>6</td><td>7</td></tr><tr><td>Y</td><td>10</td><td>12</td><td>9</td><td>11</td><td>13</td></tr></table>	X	1	3	5	6	7	Y	10	12	9	11	13	5 M
X	1	3	5	6	7									
Y	10	12	9	11	13									

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